Reg. No. | | | | | | | | | |

Question Paper Code : 12806

B.E / B.TECH DEGREE EXAMINATION, MAY / JUNE 2017

Third Semester

B.E – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What is the purpose of guard bits used in floating point operations?
2. Compare Ripple carry and carry look ahead adder.
3. Represent pictorially how Program Counter will be updated.
4. List the steps to execute the machine instruction ADD LOCA,R1.
5. What is the use of forwarding in pipelined architecture?
6. When do you say side effect occurs when executing an instruction?
7. Distinguish between cache and virtual memory.
8. Compare and contrast spatial and temporal locality of reference.
9. List the purpose of various registers available in I/O devices.
10. What is the use of PCI initialization software?

PART B – (5 × 16 = 80 Marks)

11. a) i. Write the sequence of actions needed to execute the following instructions.

- Add (immediate) number N to register R1.
- Load the contents of memory location NUM to register R1. (8 Marks)

11. a) ii. Discuss in detail about the design of Full Adder. (8 Marks)

OR

11. b) i. Write the sequence of actions needed to execute the following instructions.

- Add the contents of the memory location NUM to register R1.
- Store the content of register R1 in the location LOC. (8 Marks)

11. b) ii. Compute 23/8 using non restoring division algorithm. (8 Marks)

12. a) i. What is the need for control signal? Explain about control signals for Data path, Processor-memory interface, Instruction address generator and IR control signals. (8 Marks)

12. a) ii. Identify the sequence of steps needed to fetch and execute the following instructions:

Store R2, X(R3)

Add X (R8), R9 (8 Marks)

OR

12. b) i. Explain how hardwired control mechanism helps the processor to generate the control signals. (8 Marks)

12. b) ii. Write down the actions needed to fetch and execute the following instruction:

Load R2, 100(R1) (8 Marks)

13. a) i. List out the ways to reduce the stall in the pipeline due to branch instruction. Explain various branch prediction techniques. (8 Marks)

13. a) ii. For the following sequence of instruction, draw the pipelined execution of instructions when forwarding mechanism is followed and not followed.

ADD R1, R2, R3

SUB R4, R1, R5

MUL R6, R7, R8 (8 Marks)

OR

13. b) i. Explain with suitable example, how data dependencies are handled in 5-stage pipeline. (8 Marks)

13. b) ii. Consider the following instructions at the given addresses in the memory:

1000 Add R3, R2, #20

1004 Subtract R5, R4, #3

1008 And R6, R4, #0x3A

1012 Add R7, R2, R4

Initially, registers R2 and R4 contain 2000 and 50, respectively. These instructions are executed in a computer that has a five-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles.

(A) Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8.

(B) Describe the contents of registers IR, PC, RA, RB, RY, and RZ in the pipeline during cycles 2 to 8. (8 Marks)

14. a) i. Explain the role of TLB in virtual memory organization. (8 Marks)

14. a) ii. Consider a 2 level cache with access time of 5 ns and 80 ns and hit rates 95% and 75% respectively and memory access time is 250 ns. What is the average access time? (8 Marks)

OR

14. b) i. Explain the use of DMA controllers in a computer system with neat diagram. (8 Marks)

14. b) ii. A block-set-associative cache consists of a total of 64 blocks divided into 4-block sets. The main memory contains 4096 blocks, each consisting of 128 words.

(A) How many bits are there in a main memory address?

(B) How many bits are there in each of the TAG, SET and WORD fields?

Also, explain the direct and associative mapping techniques for the above memory specifications. (8 Marks)

15. a) i. Consider a task that reads characters typed on a keyboard, stores these data in the memory, and displays the same characters on a display screen. Show how program-controlled I/O helps in completing this task with suitable diagrams. (8 Marks)

15. a) ii. Explain in detail about PCI and explain with working principle. (8 Marks)

OR

15. b) i. Explain the Bus structure with a neat diagram. (8 Marks)

15. b) ii. What is serial interface? Explain in detail about Asynchronous serial transmission. (8 Marks)